

AUSMUN Final Project Report

Project Name: AUSMUN Management System

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Abstract

Model United Nations (MUN) conferences involve extensive logistical coordination, producing vast streams of interrelated administrative data, including delegate registrations, country representations, committee assignments, session attendance, position paper submissions, chair evaluations, and award distributions. Traditional manual tracking or spreadsheet-based records often result in fragmented data, manual cross-referencing bottlenecks, and a high vulnerability to error as the conference scales. To resolve these inefficiencies, this project presents the design and implementation of the *AUSMUN Management System*, a centralized, relationally sound database tailored to support the administration of conferences hosted at the American University of Sharjah.

Developed following the standard database design lifecycle, the core architecture translates a conceptual Entity-Relationship (ER) model comprising eight strong entities and one weak entity into a highly structured 14-table relational schema. The model captures specialized domain constraints by employing a ternary relationship to enforce committee-specific delegate evaluations, alongside composite, multivalued, and derived attributes. The database is physically realized as an Oracle SQL schema embedded with referential integrity, domain constraints, and a database view to compute derived performance metrics. System validity and performance were successfully evaluated using comprehensive mock data and a diverse set of multi-table verification queries. Finally, a completed Graphical User Interface (GUI) bridges the backend relational database with the frontend end-user, providing conference organizers and chairs with an intuitive, efficient, and reliable application to streamline large-scale event operations.

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1. AUSMUN Management System

Model United Nations (MUN) conferences bring together large numbers of participants across multiple committees, each generating its own stream of administrative data: delegate registrations, country assignments, committee schedules, session attendance, position paper submissions, chair evaluations, and award distributions. When managed manually or through disconnected spreadsheets, this volume of interrelated information quickly becomes difficult to track consistently, verify for accuracy, and query for insight, particularly as a conference scales in size or duration. A centralized, relationally sound database offers a more reliable alternative, ensuring that every piece of conference data is captured once, associated correctly with the entities it describes, and retrievable through structured queries rather than manual cross-referencing.

This report presents the design and implementation of the AUSMUN Management System, a relational database built to support the administrative needs of a Model United Nations conference hosted at the American University of Sharjah. The system is intended to help conference organizers manage delegate registration, country representation, committee assignments, session attendance, position paper submissions, chair evaluations, and award distribution within a single, coherent data model, replacing the fragmented record-keeping that typically accompanies conferences of this kind.

The project was developed for *CMP 320 – Database Systems* and follows the standard database design lifecycle: beginning with a conceptual Entity-Relationship (ER) model of the conference domain, proceeding through a systematic ER-to-relational mapping that translates entities, relationships, and special attribute types (multivalued, composite, and derived) into relational tables, continuing with the creation of a full physical implementation of that design as a working Oracle SQL schema, and then concluding with a GUI that allows end-users to interact with the database. The design was required to include a minimum of five strong entities and one weak entity; at least one 1:1, two 1:N, and two M:N relationships, along with one ternary relationship; and at least two multivalued, two composite, and two derived attributes, all of which are present in the final model and are identified explicitly throughout this report.

Beyond the conceptual and relational design, this report also documents the transition from paper design to a functioning database: the complete Oracle DDL for all fourteen tables, the additional constraints introduced during implementation to enforce rules that the original design had left undefined, a derived-attribute view for evaluation scoring, representative sample data, and a set of test queries used to validate that the schema behaves as intended across every relationship type.

2. Project Description

The database system is an **AUSMUN Management System**. The system keeps track of the main information needed to organize and manage a Model United Nations conference. It stores data about **delegates, committees, countries, chairs, sessions, attendance, position papers, evaluations, and awards**. The purpose of the system is to help conference organizers manage delegate registration, country representation, committee assignments, session attendance, position paper submissions, chair evaluations, and award distribution in an organized and efficient way.

The **Delegate** entity stores information about each participant in the conference. Each delegate is uniquely identified by **DelegateID**. The main attributes of a delegate are **DelegateID, Name, Email, School, DateOfBirth, Age, and RegistrationStatus**. The delegate name is a **composite attribute** that can be divided into **FirstName** and **LastName**. The **Age** attribute is a **derived attribute** because it can be calculated from the delegate's date of birth. Each delegate represents one country and is assigned to one committee. A delegate may submit position papers, attend sessions through attendance records, receive evaluations from chairs, and possibly receive awards.

The **Workshop** entity stores information about optional training seminars and guest speaker events offered during the conference. Each workshop is uniquely identified by **WorkshopID**. The main attributes of a workshop are **WorkshopID, WorkshopTitle, SpeakerName, and RoomNumber**. Since a single delegate can register for and attend multiple workshops, and each workshop can be attended by many delegates, the relationship between Delegate and Workshop is many-to-many (M:N).

The **Country** entity stores information about the countries represented in AUSMUN. Each country is uniquely identified by **CountryID**. The main attributes of a country are **CountryID, CountryName, Region, and VetoPower**. One country may be represented by many delegates, while each delegate represents only one country. Therefore, the relationship between **Country** and **Delegate** is **one-to-many**. Participation is **total on the Delegate side** because every delegate must represent a country, while it is **partial on the Country side** because not every country must necessarily have a delegate.

The **Committee** entity stores information about each committee in the conference. Each committee is uniquely identified by **CommitteeID**. The main attributes of a

committee are **CommitteeID**, **CommitteeName**, **CommitteeType**, **BuildingName**, **RoomNumber**, and **Topic**. The **Topic** attribute is shown as a **multivalued attribute** because a committee may discuss more than one topic. Each committee can have many delegates assigned to it, while each delegate is assigned to only one committee. Therefore, the relationship between **Committee** and **Delegate** is **one-to-many**. A committee can also have many sessions, many position papers, many evaluations, and many awards connected to it.

The **Chair** entity stores information about the chairs responsible for supervising committees and evaluating delegates. Each chair is uniquely identified by **ChairID**. The main attributes of a chair are **ChairID**, **ChairName**, **ChairEmail**, and **AssignedTopic**. The chair name is a **composite attribute** that can be divided into **FirstName** and **LastName**. A committee may be supervised by many chairs, while each chair supervises one committee, meaning that the relationship between committee & chair is **one-to-many**. The chair also participates in the **Evaluates** relationship, where a chair evaluates a delegate in a specific committee.

The **Session** entity stores information about the sessions held during the conference. Each session is uniquely identified by **SessionID**. The main attributes of a session are **SessionID**, **SessionDate**, **StartTime**, and **EndTime**. Each session is connected to attendance records. Since many delegates can attend many sessions, the relationship between **Delegate** and **Session** is a **many-to-many relationship**. This many-to-many relationship is solved using the **Attendance** entity.

The **Attendance** entity records whether delegates attended conference sessions. Each attendance record is uniquely identified by its associated **DelegateID**, and **SessionID**. Attendance acts as an associative entity between **Delegate** and **Session**. One delegate can have many attendance records, and one session can have many attendance records. Therefore, the relationship becomes **Delegate 1:N Attendance** and **Session 1:N Attendance**. Attendance has **total participation** because every attendance record must belong to one delegate and one session.

The **Position_Paper** entity stores information about the papers submitted by delegates. A position paper is identified by a **PaperNumber**. The main attributes of a position paper are **PaperNumber**, **SubmissionDate**, **Content**, **Score**, **Topic**, **DelegateID**, and **CommitteeID**. A delegate can submit one position paper and each position paper is submitted by one delegate. Therefore, the relationship between **Delegate** and **Position_Paper** is **one-to-one**. Each position paper is also written for one specific committee, while one committee can have many position papers. Therefore,

the relationship between **Committee** and **Position_Paper** is also **one-to-many**. The **Position_Paper** entity has total participation with Delegate and Committee because a paper cannot exist without the delegate who submitted it & the committee it belongs to.

The **Evaluates** relationship stores the performance evaluation given by chairs to delegates. It is a **ternary relationship** because it depends on three entities: **Chair**, **Delegate**, and **Committee**, because a chair evaluates a delegate based on the specific academic standards and context of a given committee. Furthermore, it ensures that chairs grade delegates within their committee. The main attributes of the evaluation are **ChairID**, **DelegateID**, **CommitteeID**, **EvaluationScore**, and **Feedback**. A chair evaluates a delegate within a specific committee. One chair can write many evaluations, one delegate can receive many evaluations, and one committee can have many evaluations. Therefore, in relational mapping, this relationship can be converted into an **Evaluation table** connected to Chair, Delegate, and Committee nevertheless **EvaluationScore** will have to be derived from **Feedback** that is a multivalued composite attribute that contains professionalism, research, and public speaking.

The **Awards** entity stores information about awards given to delegates based on their performance. Each award is uniquely identified by **AwardID**. The main attributes of an award are **AwardID**, **AwardTitle**, **AwardDate**, **DelegateID**, and **CommitteeID**. A delegate may receive many awards, and each award can be received by multiple delegates (i.e: honorable mentions, commendations, etc). Therefore, the relationship between **Delegate** and **Awards** is **many-to-many**. A committee may give many awards, but each award is given by one committee. Therefore, the relationship between **Committee** and **Awards** is **one-to-many**. Awards are optional for delegates because not every delegate receives an award.

The system also includes special attribute types. The **Delegate Name** and **Chair Name** are **composite attributes** because they can be divided into first name and last name. The **Delegate Age** is a **derived attribute** because it can be calculated from DateOfBirth. The **Committee Topic** is a **multivalued attribute** because a committee may have more than one topic. The **Evaluation Feedback** is a multivalued composite attribute because a chair may have more than one piece of feedback to give a delegate.

3. ER Modeling

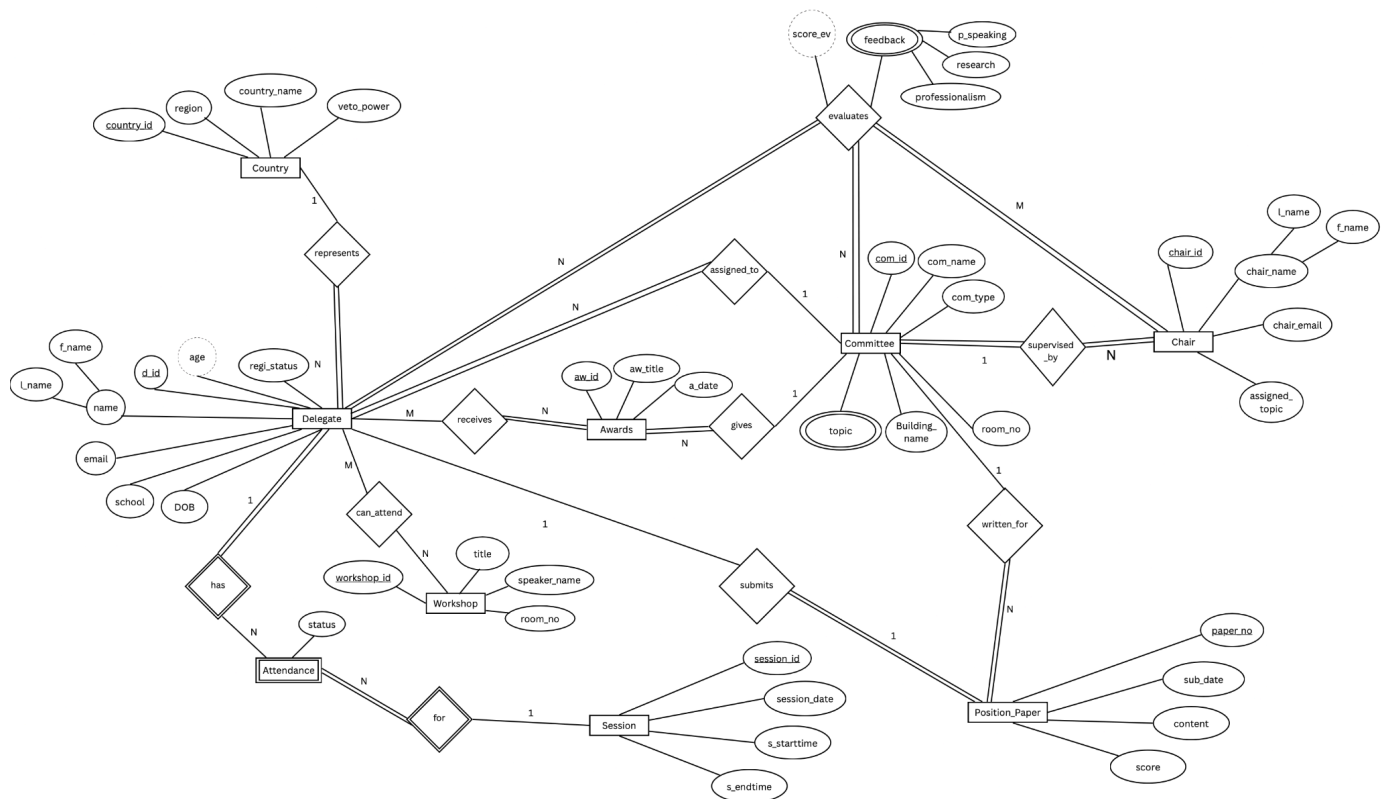
In this section, we provided the Entity-Relationship (ER) diagram. Per the project requirements, the design includes at least five strong entities and one weak entity. Other component details include:

- **Entities:** (Minimum 5 strong, 1 weak)
- **Relationships:** (Minimum 1:1, two 1:M, two M:N, and one ternary)
- **Attributes:** (Minimum 2 multi-valued, 2 composite, and 2 derived)

3.1 ER Diagram

A link to our ER Diagram is provided for a better visual experience if necessary:

<https://canva.link/mvl8llc5x9c4416>



3.2 Design Rationale for the ER Diagram

The ER diagram was built around the core real-world objects a MUN conference organizer needs to track, and each entity, relationship, and attribute type was chosen deliberately to satisfy both the practical needs of the system and the project's structural requirements (minimum entity counts, relationship cardinalities, and attribute types).

Entity choices

Delegate, Committee, Country, Chair, Session, Position_Paper, Awards, and Workshop were modeled as strong entities because each has its own independent existence and a natural, self-contained identifier (DelegateID, CommitteeID, CountryID, ChairID, SessionID, PaperNumber, AwardID, WorkshopID). Attendance was modeled as a weak entity rather than a strong one because an attendance record has no independent identity of its own — it only makes sense in the context of a specific delegate at a specific session, and it exists solely to resolve the many-to-many relationship between Delegate and Session. This is why Attendance's identifier is a composite of DelegateID and SessionID rather than a surrogate key of its own.

Relationship cardinalities and participation

Each relationship's cardinality reflects a real constraint on how the conference actually runs. Country–Delegate and Committee–Delegate are one-to-many because a delegate can only ever represent one country and sit on one committee at a time, while a country or committee can naturally host many delegates. Participation is total on the Delegate side of both relationships (a delegate cannot exist in the system without a country and a committee) but partial on the Country and Committee sides (a country need not have a delegate registered, and in principle a committee could exist before delegates are assigned to it). Committee–Chair is one-to-many for a comparable reason: a chair supervises exactly one committee, but a committee may be supervised by several chairs, since MUN committees are typically staffed by a small panel rather than a single person.

Delegate–Position_Paper is one-to-one because each delegate submits exactly one position paper, and each paper belongs to exactly one delegate; participation is total on both sides, since neither a delegate's registration nor a position paper is considered complete without the other. Committee–Position_Paper is one-to-many, since a committee receives many papers but each paper is written for only one committee.

Delegate–Session is many-to-many, since a delegate attends multiple sessions over the course of a conference and a session is attended by many delegates simultaneously; the Attendance weak entity resolves this into two one-to-many relationships (Delegate HAS Attendance, Attendance FOR Session) with total participation on the Attendance side, since an attendance record cannot exist without both a delegate and a session behind it. Delegate–Workshop and Delegate–Awards are both many-to-many for similar reasons: a delegate can attend several optional workshops and a workshop hosts many delegates, and a delegate can win more than one award (or share an award with others, as with honorable mentions) while a committee gives out multiple awards over the conference.

Evaluations were modeled as a three-way relationship between Chair, Delegate, and Committee rather than as two separate binary relationships (Chair–Delegate and Delegate–Committee) because an evaluation only makes sense as a single combined fact: a specific chair grading a specific delegate within the context of a specific committee. Collapsing this into binary relationships would lose the constraint that a chair should only evaluate delegates within their own assigned committee, a rule we wanted the model itself to express.

Special attribute types

We used composite attributes for Delegate Name and Chair Name because both are naturally split into first and last name for storage and querying purposes, but conceptually represent a single "name" idea. Age was modeled as a derived attribute rather than a stored one because it can always be recalculated from DateOfBirth, and storing it directly would risk it becoming stale or inconsistent over time. The same logic applies to EvaluationScore, which we derived from the Feedback components (professionalism, research, and public speaking) rather than storing it as an independent value. Topic was modeled as a multivalued attribute of Committee because a single committee frequently debates more than one topic over the course of a conference, and Feedback was modeled as a multivalued composite attribute of the Evaluates relationship because a chair may leave more than one piece of structured feedback (each itself composed of the three scoring components) for the same delegate.

Taken together, these choices ensure the diagram satisfies the project's minimum requirements while also reflecting decisions we made deliberately about how the conference itself operates.

4. ER to Relational Mapping

Step 1: Mapping of Regular [Strong] Entity Types

First, we map the strong entities within the ER Diagram with their simple attributes, omitting any derived and/or multivalued attributes, and flattening composite attributes. Foreign keys are not included yet.

Strong entities within the schema are:

- Country (country_id (PK), region, country_name, veto_power)
Primary Key {country_id}
- Delegate (d_id (PK), f_name, l_name, email, school, DOB, regi_status)
 - [we do NOT add age because it is a derived attribute of DOB, and we flatten “name” to f_name and l_name respectively]Primary Key {d_id}
- Awards (aw_id (PK), aw_title, a_date)
Primary Key {aw_id}
- Workshop (workshop_id (PK), title, speaker_name, room_no)
Primary Key {workshop_id}
- Session (session_id (PK), session_date, s_starttime, s_endtime)
Primary Key {session_id}
- Committee (com_id (PK), com_name, com_type, building_name, room_no)
 - [we do NOT add topic yet as it is multivalued]Primary Key {com_id}
- Position_Paper (paper_no (PK), sub_date, content, score)
Primary Key {paper_no}
- Chair (chair_id (PK), f_name, l_name, chair_email, assigned_topic)
Primary Key {chair_id}

Step 2: Mapping of Weak Entity Types

We then map the weak entities within the ER diagram with their simple attributes, including the foreign key attributes of the strong entities they are dependent upon.

Weak entities within the schema are:

- Attendance (status, d_id (PK) (FK), session_id (PK) (FK))
Primary Key {d_id, session_id}

Step 3: Mapping of 1:1 Relationship Types

We then map all 1:1 relationships and, depending on participation, we either create a foreign key in one of the relations, we merge the relations, or we create a third relation. The 1:1 relationships and the corresponding actions are:

- Delegate SUBMITS Position_Paper
 - Delegate's primary key d_id will be included as a foreign key in Position_Paper, however, since Position_Paper is a strong entity, d_id will not be used as a primary key. Position_Paper is now updated as:
Position_Paper (paper_no (PK), sub_date, content, score, d_id (FK))

Step 4: Mapping of 1:N Relationship Types

We then map all 1:N relationships, including the primary key of the entity on the 1-side of the relationship as a foreign key in the entity on the n-side of the relationship. We also include any simple attributes of the 1:N relationship as attributes of the entity on the n-side.

The 1:N relationships and the corresponding actions are:

- Delegate HAS Attendance
 - Delegate's primary key d_id will be included as a foreign key AND primary key in Attendance as Attendance is a weak entity and HAS is an identifying relationship. This was established when we mapped the weak entities.
- Attendance FOR Session
 - Session's primary key session_id will be included as a foreign key AND primary key in Attendance as Attendance is a weak entity and FOR is an identifying relationship. This was established when we mapped the weak entities.
- Delegate REPRESENTS Country
 - Country's primary key country_id will be included as a foreign key in Delegate. Delegate will now be updated as follows:
Delegate (d_id (PK), f_name, l_name, email, school, DOB, regi_status, country_id (FK))
- Delegate ASSIGNED TO Committee
 - Committee's primary key com_id will be included as a foreign key in Delegate. Delegate will now be updated as follows:
Delegate (d_id (PK), f_name, l_name, email, school, DOB, regi_status, country_id (FK), com_id (FK))
- Committee GIVES Awards

- Committee's primary key `com_id` will be included as a foreign key in Awards. Award will now be updated as follows:
Awards (`aw_id` (PK), `aw_title`, `a_date`, `com_id` (FK))
- Committee SUPERVISED BY Chair
 - Committee's primary key `com_id` will be included as a foreign key in Chair. Chair will now be updated as follows:
Chair (`chair_id` (PK), `f_name`, `l_name`, `chair_email`, `assigned_topic`, `com_id` (FK))
- Position_Paper WRITTEN FOR Committee
 - Committee's primary key `com_id` will be included as a foreign key in Position_Paper. Position_Paper will now be updated as follows:
Position_Paper (`paper_no` (PK), `sub_date`, `content`, `score`, `d_id` (FK), `com_id` (FK))

Step 5: Mapping of M:N Relationship Types

We then map all the M:N relationships, creating a new relation to represent the relationship. We include in the relation the primary keys of the entities involved as foreign key attributes. We also include any simple attributes of the relationship.

The M:N relationships and the corresponding actions are:

- Delegate RECEIVES Awards
 - We will create a new table AWARD_RECIPIENTS that contains the primary key of Delegate `d_id` and the primary key of Awards `aw_id` as foreign keys AND primary keys. AWARD_RECIPIENTS is then as follows:

AWARD_RECIPIENTS (`d_id` (PK) (FK), `aw_id` (PK) (FK))
Primary Keys {`d_id` (FK), `aw_id` (FK)}
- Delegate CAN ATTEND Workshop
 - We will create a new table WORKSHOP_ATTENDEES that contains the primary key of Delegate `d_id` and the primary key of Workshop `workshop_id` as foreign AND primary keys. WORKSHOP_ATTENDEES is then as follows:

WORKSHOP_ATTENDEES (`d_id` (PK) (FK), `workshop_id` (PK) (FK))
Primary Keys {`d_id` (FK), `workshop_id` (FK)}

Step 6: Mapping of N-ary Relationship Types

We then map all n-ary relationships, creating a new relation to represent the relationship. We include in the relation the primary keys of the entities involved as foreign key attributes. We also include any simple attributes of the relationship.

The n-ary relationships and the corresponding actions are:

- EVALUATES between Delegate, Chair, and Committee
 - We will create a new table EVALUATIONS that contains the primary keys d_id (Delegate), chair_id (Chair), and com_id (Committee) as foreign AND primary keys. We will not include feedback as it is a multivalued attribute. EVALUATIONS is then as follows:

EVALUATIONS (d_id (PK) (FK), chair_id (PK) (FK), com_id (PK) (FK))
Primary Keys {d_id (FK), chair_id (FK), com_id (FK)}

Step 7: Mapping of Multivalued Attributes

We then map all multivalued attributes, creating a new relation to represent the attribute A. We include in the relation the corresponding attributes, and the primary keys K of the entities involved as foreign key attributes. The primary key of the relation is a combination of A and K.

The multivalued attributes and the corresponding actions are:

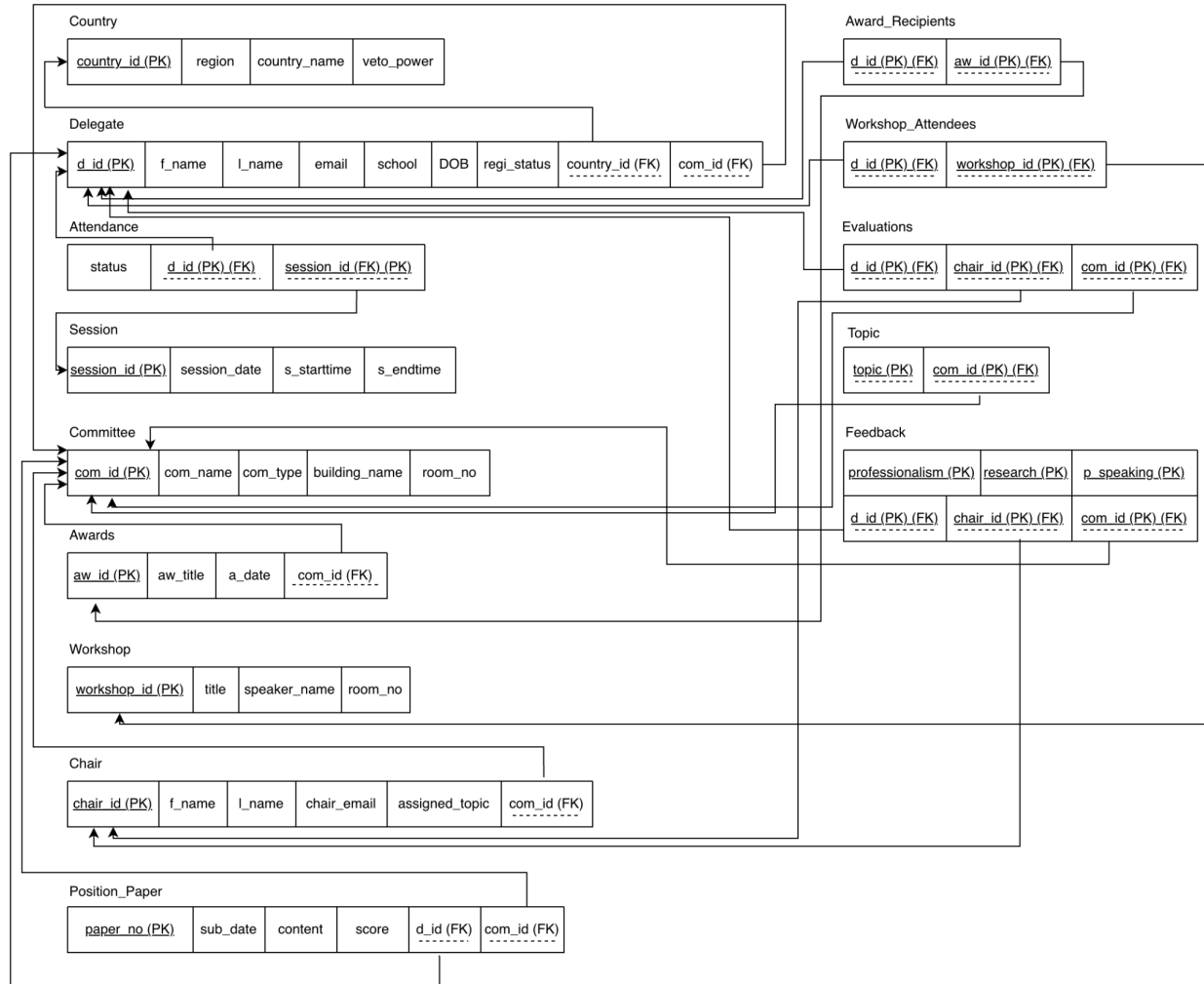
- TOPIC as a multivalued attribute of Committee
 - We will create a new table TOPIC that contains an attribute topic and that contains the primary key of Committee com_id. These two attributes will be the combined primary key of TOPIC. TOPIC is then as follows:

TOPIC (topic (PK), com_id (PK) (FK))
Primary Keys {topic, com_id (FK)}

- FEEDBACK as a multivalued attribute of Evaluations
 - We will create a new table FEEDBACK that contains attributes professionalism, research, and p_speaking. However, since we just created Evaluations as a weak relation, FEEDBACK must inherit all of its primary keys, being d_id (Delegate), chair_id (Chair), and com_id (Committee) as foreign AND primary keys. These keys and attributes will act as FEEDBACK's composite primary key. Thus FEEDBACK is as follows:

FEEDBACK (d_id (PK) (FK), chair_id (PK) (FK), com_id (PK) (FK),
professionalism (PK), research (PK), p_speaking (PK))
Primary Keys {d_id (FK), chair_id (FK), com_id (FK), professionalism,
research, p_speaking}

The relational map then looks like this:



5. Keys and Constraints

5.1 Keys:

Entity	Primary Key (PK)	Foreign Keys (FK)
Delegate	d_id	country_id, com_id
Committee	com_id	none
Country	country_id	none
Chair	chair_id	com_id
Session	session_id	none
Attendance	d_id + session_id (comp.)	d_id, session_id
Position_Paper	paper_no	d_id, com_id
Evaluation	chair_id + d_id + com_id (comp.)	chair_id, d_id, com_id
Award	aw_id	com_id
Workshop	workshop_id	none
Workshop_Attendees	d_id, workshop_id (comp.)	d_id, workshop_id
Topic	topic, com_id (Composite)	com_id
Award_Recipients	d_id, aw_id (comp.)	d_id, aw_id
Feedback	d_id, chair_id, com_id, professionalism, research, p_speaking (comp.)	d_id, chair_id, com_id

5.2 Constraints:

Entity	Domain Constraints	Mapping Constraints
Delegate	Age is a derived attribute calculated from DateOfBirth. country_id & com_id must be NOT NULL to indicate full participation. f_name, l_name, email, and DOB also NOT NULL for data integrity.	1:N (Many-to-One): Represents exactly one Country (Total participation). 1:N (Many-to-One): Assigned to exactly one Committee. 1:N: One student has many Attendance records. M:N (Many-to-Many): Registers for many Workshops (via Workshop_Attendees), and can receive multiple Awards (via Award_Recipients).
Committee	com_name & com_type NOT NULL for data integrity. com_name UNIQUE to prevent duplicate committee names.	1:M (One-to-Many): Associated with many Delegates, Position_Papers, and Awards. 1:N: Supervised by many Chairs. Ternary: Is used by a chair to evaluate a delegate; many chairs can use the committee to evaluate many delegates Multivalued: Topic must be mapped to a separate table.
Country	country_region & country_name NOT NULL for data integrity. country_name UNIQUE to prevent duplicate countries.	1:N (One-to-Many): One country can be represented by many delegates.
Chair	com_id must be set to NOT NULL to indicate full participation f_name, l_name, and email also NOT NULL for data integrity email UNIQUE to prevent re-registration of chairs	1:N (Many-to-One): Assigned to one committee . Ternary: Evaluates delegates within a specific committee.
Session	session_date, s_starttime, s_endtime NOT NULL for data integrity	1:N (One-to-Many): One session has many Attendance records.

Attendance	status is restricted to specific values: present, present_and_voting, absent, or late.	1:N (One-to-Many): Each Attendance record is associated with a Delegate, and with a Session. Associative Entity: Resolves the M:N relationship between Delegate and Session.
Position_Paper	score must follow a valid numeric domain. d_id & com_id must be NOT NULL to indicate total participation d_id UNIQUE to ensure that each Position_Paper is only associated with ONE Delegate	Strong Entity with Total Participation: Linked to Delegate via a 1:1 relationship and to Committee via a 1:N relationship. Dependent on both for existence but not identity.
Evaluation	EvaluationScore is a derived attribute calculated from Feedback components (professionalism, research, and public speaking).	Ternary Relationship: Represents a 3-way dependency between Chair, Delegate, and Committee.
Award	com_id must be NOT NULL to indicate total participation aw_title NOT NULL for data integrity.	1:M (One-to-Many): Given by exactly one Committee. M:N (Many-to-Many): Can be received by multiple Delegates (resolved via Award_Recipients table). Optional for delegates.
Workshop	workshop_title NOT NULL for data integrity.	M:N (Many-to-Many): Hosted for many Delegates (resolved via Workshop_Attendees table).
Workshop_Attendees	Composite key domain restrictions.	Bridge Table: Resolves the M:N relationship between Delegate and Workshop.
Award_Recipients	Composite key domain restrictions.	Multivalued Attribute Table: Resolves the multi-valued Topic attribute owned by the Committee entity.
Topic	Simple text domain for topic values.	Multivalued Attribute Table: Resolves the multi-valued Topic attribute owned by the Committee entity.
Feedback	Contains individual component attributes: professionalism, research, and p_speaking.	Multivalued Composite Attribute Table: Maps the multiple evaluation performance tracking dimensions back to the Evaluation entity.

6. Physical Implementation in Oracle SQL

The logical design above was implemented as a live Oracle schema of 14 tables plus one view, seeded with representative sample data and exercised with 12 test queries. This section documents the DDL, the constraints added beyond what the original proposal specified, and the design corrections made along the way.

Session was renamed to mun_session. SESSION is a reserved word in Oracle SQL, so the entity was implemented as mun_session. This is a naming change only; the attributes and relationships are unchanged from the proposal.

6.1. Constraints Added During Implementation

Table	Constraint Added	Rationale
country	veto_power CHECK IN ('Y','N'), DEFAULT 'N'	Restricts the veto-power flag to a boolean-style domain.
delegate	regi_status CHECK IN ('Registered','Pending','Waitlisted','Cancelled'), DEFAULT 'Registered'	Defines the registration-status domain.
delegate	delegate_did_com_uk UNIQUE(d_id, com_id)	Supports the composite FK from evaluations, guaranteeing a delegate's committee assignment is traceable.
chair	chair_email_uk UNIQUE(chair_email)	Enforces one account per chair email.
chair	chair_id_com_uk UNIQUE(chair_id, com_id)	Supports the composite FK from evaluations, ensuring a chair evaluates only within their own committee.
position_paper	pp_score_ck CHECK (score BETWEEN 0 AND 100)	Fills the “valid numeric domain” requirement noted in the proposal.
position_paper	pp_delegate_com_fk FOREIGN KEY (d_id, com_id) REFERENCES delegate(d_id, com_id)	New business rule: a delegate's position paper must be written for that same delegate's assigned committee.
attendance	attendance_status_ck CHECK IN ('present','present_and_voting','absent','late')	Implements the status domain named in the proposal's constraints table.
evaluations	eval_delegate_com_fk / eval_chair_com_fk (composite FKs)	Implements the proposal's own requirement that “chairs grade delegates within their committee.”

Table	Constraint Added	Rationale
feedback	feedback_prof_ck / feedback_research_ck / feedback_speaking_ck CHECK (BETWEEN 0 AND 10)	Bounds each Feedback component to a 0–10 scale.

6.2. Table DDL

Tables were created in dependency order (referenced tables before their foreign-key dependent tables).

country

```
CREATE TABLE country (
    country_id    NUMBER(4),
    region        VARCHAR2(50) CONSTRAINT country_region_nn NOT NULL,
    country_name  VARCHAR2(80) CONSTRAINT country_name_nn NOT NULL,
    veto_power    CHAR(1) DEFAULT 'N' CONSTRAINT country_veto_nn NOT NULL,

    CONSTRAINT country_pk PRIMARY KEY (country_id),
    CONSTRAINT country_name_uk UNIQUE (country_name),
    CONSTRAINT country_veto_ck CHECK (veto_power IN ('Y', 'N'))
);
```

committee

```
CREATE TABLE committee (
    com_id        NUMBER(4),
    com_name      VARCHAR2(100) CONSTRAINT committee_name_nn NOT NULL,
    com_type      VARCHAR2(50) CONSTRAINT committee_type_nn NOT NULL,
    building_name VARCHAR2(80),
    room_no       VARCHAR2(20),

    CONSTRAINT committee_pk PRIMARY KEY (com_id),
    CONSTRAINT committee_name_uk UNIQUE (com_name)
);
```

workshop

```
CREATE TABLE workshop (
    workshop_id   NUMBER(4),
    title         VARCHAR2(120) CONSTRAINT workshop_title_nn NOT NULL,
    speaker_name  VARCHAR2(100),
    room_no       VARCHAR2(20),

    CONSTRAINT workshop_pk PRIMARY KEY (workshop_id)
);
```

mun_session

```
CREATE TABLE mun_session (  
    session_id      NUMBER(4),  
    session_date    DATE CONSTRAINT session_date_nn NOT NULL,  
    s_starttime     VARCHAR2(5) CONSTRAINT session_start_nn NOT NULL,  
    s_endtime       VARCHAR2(5) CONSTRAINT session_end_nn NOT NULL,  
  
    CONSTRAINT mun_session_pk PRIMARY KEY (session_id)  
);
```

delegate

```
CREATE TABLE delegate (  
    d_id            NUMBER(4),  
    f_name          VARCHAR2(50) CONSTRAINT delegate_fname_nn NOT NULL,  
    l_name          VARCHAR2(50) CONSTRAINT delegate_lname_nn NOT NULL,  
    email           VARCHAR2(120) CONSTRAINT delegate_email_nn NOT NULL,  
    school          VARCHAR2(120),  
    dob             DATE CONSTRAINT delegate_dob_nn NOT NULL,  
    regi_status     VARCHAR2(20) DEFAULT 'Registered' CONSTRAINT delegate_status_nn NOT  
NULL,  
    country_id      NUMBER(4) CONSTRAINT delegate_country_nn NOT NULL,  
    com_id          NUMBER(4) CONSTRAINT delegate_committee_nn NOT NULL,  
  
    CONSTRAINT delegate_pk PRIMARY KEY (d_id),  
    CONSTRAINT delegate_email_uk UNIQUE (email),  
    CONSTRAINT delegate_did_com_uk UNIQUE (d_id, com_id),  
    CONSTRAINT delegate_status_ck CHECK (regi_status IN ('Registered', 'Pending',  
'Waitlisted', 'Cancelled')),  
    CONSTRAINT delegate_country_fk FOREIGN KEY (country_id) REFERENCES  
country(country_id),  
    CONSTRAINT delegate_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id)  
);
```

chair

```
CREATE TABLE chair (  
    chair_id        NUMBER(4),  
    f_name          VARCHAR2(50) CONSTRAINT chair_fname_nn NOT NULL,  
    l_name          VARCHAR2(50) CONSTRAINT chair_lname_nn NOT NULL,  
    chair_email     VARCHAR2(120) CONSTRAINT chair_email_nn NOT NULL,  
    assigned_topic  VARCHAR2(150),  
    com_id          NUMBER(4) CONSTRAINT chair_committee_nn NOT NULL,  
  
    CONSTRAINT chair_pk PRIMARY KEY (chair_id),  
    CONSTRAINT chair_email_uk UNIQUE (chair_email),  
    CONSTRAINT chair_id_com_uk UNIQUE (chair_id, com_id),
```

```

CONSTRAINT chair_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id)
);

```

awards

```

CREATE TABLE awards (
  aw_id      NUMBER(4),
  aw_title   VARCHAR2(100) CONSTRAINT awards_title_nn NOT NULL,
  a_date     DATE,
  com_id     NUMBER(4) CONSTRAINT awards_committee_nn NOT NULL,

  CONSTRAINT awards_pk PRIMARY KEY (aw_id),
  CONSTRAINT awards_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id)
);

```

position_paper

```

CREATE TABLE position_paper (
  paper_no   NUMBER(4),
  sub_date   DATE,
  content     VARCHAR2(2000),
  score      NUMBER(5,2),
  d_id       NUMBER(4) CONSTRAINT pp_delegate_nn NOT NULL,
  com_id     NUMBER(4) CONSTRAINT pp_committee_nn NOT NULL,

  CONSTRAINT position_paper_pk PRIMARY KEY (paper_no),
  CONSTRAINT pp_score_ck CHECK (score BETWEEN 0 AND 100),
  CONSTRAINT pp_delegate_uk UNIQUE (d_id),
  CONSTRAINT pp_delegate_fk FOREIGN KEY (d_id) REFERENCES delegate(d_id),
  CONSTRAINT pp_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id),
  CONSTRAINT pp_delegate_com_fk FOREIGN KEY (d_id, com_id) REFERENCES delegate(d_id,
com_id)
);

```

attendance

```

CREATE TABLE attendance (
  d_id       NUMBER(4),
  session_id NUMBER(4),
  status     VARCHAR2(20) CONSTRAINT attendance_status_nn NOT NULL,

  CONSTRAINT attendance_pk PRIMARY KEY (d_id, session_id),
  CONSTRAINT attendance_status_ck CHECK (status IN ('present', 'present_and_voting',
'absent', 'late')),
  CONSTRAINT attendance_delegate_fk FOREIGN KEY (d_id) REFERENCES delegate(d_id),
  CONSTRAINT attendance_session_fk FOREIGN KEY (session_id) REFERENCES
mun_session(session_id)
);

```

workshop_attendees

```
CREATE TABLE workshop_attendees (  
    d_id          NUMBER(4),  
    workshop_id   NUMBER(4),  
  
    CONSTRAINT workshop_attendees_pk PRIMARY KEY (d_id, workshop_id),  
    CONSTRAINT wa_delegate_fk FOREIGN KEY (d_id) REFERENCES delegate(d_id),  
    CONSTRAINT wa_workshop_fk FOREIGN KEY (workshop_id) REFERENCES  
workshop(workshop_id)  
);
```

award_recipients

```
CREATE TABLE award_recipients (  
    d_id   NUMBER(4),  
    aw_id  NUMBER(4),  
  
    CONSTRAINT award_recipients_pk PRIMARY KEY (d_id, aw_id),  
    CONSTRAINT ar_delegate_fk FOREIGN KEY (d_id) REFERENCES delegate(d_id),  
    CONSTRAINT ar_award_fk FOREIGN KEY (aw_id) REFERENCES awards(aw_id)  
);
```

topic

```
CREATE TABLE topic (  
    topic   VARCHAR2(150),  
    com_id  NUMBER(4),  
  
    CONSTRAINT topic_pk PRIMARY KEY (topic, com_id),  
    CONSTRAINT topic_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id)  
);
```

evaluations

```
CREATE TABLE evaluations (  
    d_id      NUMBER(4),  
    chair_id  NUMBER(4),  
    com_id    NUMBER(4),  
  
    CONSTRAINT evaluations_pk PRIMARY KEY (d_id, chair_id, com_id),  
    CONSTRAINT eval_delegate_com_fk FOREIGN KEY (d_id, com_id) REFERENCES  
delegate(d_id, com_id),  
    CONSTRAINT eval_chair_com_fk FOREIGN KEY (chair_id, com_id) REFERENCES  
chair(chair_id, com_id),  
    CONSTRAINT eval_committee_fk FOREIGN KEY (com_id) REFERENCES committee(com_id)  
);
```

feedback

```
CREATE TABLE feedback (  
    d_id            NUMBER(4),  
    chair_id        NUMBER(4),  
    com_id          NUMBER(4),  
    professionalism NUMBER(2),  
    research         NUMBER(2),  
    p_speaking      NUMBER(2),  
  
    CONSTRAINT feedback_pk PRIMARY KEY (d_id, chair_id, com_id, professionalism,  
research, p_speaking),  
    CONSTRAINT feedback_eval_fk FOREIGN KEY (d_id, chair_id, com_id) REFERENCES  
evaluations(d_id, chair_id, com_id),  
    CONSTRAINT feedback_prof_ck CHECK (professionalism BETWEEN 0 AND 10),  
    CONSTRAINT feedback_research_ck CHECK (research BETWEEN 0 AND 10),  
    CONSTRAINT feedback_speaking_ck CHECK (p_speaking BETWEEN 0 AND 10)  
);
```

6.3 Derived-Attribute View

EvaluationScore was specified in the proposal as a derived attribute (calculated from the Feedback components) rather than a stored column. It is implemented as a view over feedback, averaging professionalism, research, and p_speaking per (delegate, chair, committee):

evaluation_scores

```
CREATE VIEW evaluation_scores AS  
SELECT  
    d_id,  
    chair_id,  
    com_id,  
    ROUND(AVG((professionalism + research + p_speaking) / 3), 2) AS score_ev  
FROM feedback  
GROUP BY d_id, chair_id, com_id;
```

6.4 Sample Data

The schema was seeded with 6 countries, 3 committees, 6 topics, 3 workshops, 3 sessions, 6 delegates, 6 chairs, 4 awards, 6 position papers, 18 attendance records, 8 workshop-attendee records, 6 award-recipient records, 7 evaluations, and 7 feedback rows, enough to exercise every relationship type, including a shared award (two delegates receiving the same “Outstanding Delegate”-style award) and a delegate with more than one feedback entry from different chairs.

country

	COUNTRY_ID	REGION	COUNTRY_NAME	VETO_POWER
1	1	Middle East	United Arab Emirates	N
2	2	North America	United States of America	Y
3	3	Europe	France	Y
4	4	Africa	Egypt	N
5	5	Asia	India	N
6	6	Asia	China	Y

committee

	COM_ID	COM_NAME	COM_TYPE	BUILDING_NAME	ROOM_NO
1	10	ECOSOC	Specialized	ESB	1001
2	20	UNSC	Security Council	ESB	1002
3	30	WHO	Specialized	ESB	1003

workshop

	WORKSHOP_ID	TITLE	SPEAKER_NAME	ROOM_NO
1	501	Rules of Procedure Training	Dr. Mohamed Ibrahim	2001
2	502	Public Speaking for Delegates	Youssef Hassan	2002
3	503	Research and Position Paper Writing	Ranim Ali	2003

mun_session

	SESSION_ID	SESSION_DATE	S_STARTTIME	S_ENDTIME
1	1001	30/01/26	09:00	10:30
2	1002	30/01/26	11:00	12:30
3	1003	31/01/26	09:00	10:30

delegate

	D_ID	F_NAME	L_NAME	EMAIL	SCHOOL	DOB	REGI_STATUS	COUNTRY_ID	COM_ID
1	101	Ahmed	Mohsen	ahmed.mohsen@example.com	American University of Sharjah	15/03/05	Registered	4	10
2	102	Fedaa	Elmahdi	fedaa.elmahdi@example.com	American University of Sharjah	20/10/04	Registered	1	10
3	103	Daniel	Zakhari	daniel.zakhari@example.com	American University of Sharjah	12/01/05	Registered	2	20
4	104	Renad	Hamad	renad.hamad@example.com	American University of Sharjah	09/07/04	Registered	3	20
5	105	Sara	Khan	sara.khan@example.com	AUS High School	11/05/06	Registered	5	30
6	106	Omar	Li	omar.li@example.com	Sharjah International School	02/12/05	Registered	6	30

chair

	CHAIR_ID	F_NAME	L_NAME	CHAIR_EMAIL	ASSIGNED_TOPIC	COM_ID
1	201	Youssef	Hassan	youssef.hassan@example.com	Youth Employment	10
2	202	Ranim	Ali	ranim.ali@example.com	Labor Supply Chains	10
3	203	Shamma	Almualla	shamma.almualla@example.com	Cybersecurity	20
4	204	Reema	Shubair	reema.shubair@example.com	Peacekeeping	20
5	205	Mariam	Saeed	mariam.saeed@example.com	Pandemic Preparedness	30
6	206	Khaled	Nasser	khaled.nasser@example.com	Mental Health	30

awards

	AW_ID	AW_TITLE	A_DATE	COM_ID
1	301	Best Delegate	01/02/26	10
2	302	Outstanding Delegate	01/02/26	10
3	303	Honorable Mention	01/02/26	20
4	304	Best Position Paper	01/02/26	30

position_paper

	PAPER_NO	SUB_DATE	CONTENT	SCORE	D_ID	COM_ID
1	401	25/01/26	Position paper on youth employment and labor policy.	92	101	10
2	402	25/01/26	Position paper on labor rights and sustainable supply chains.	88	102	10
3	403	26/01/26	Position paper on cybersecurity and peacekeeping.	85	103	20
4	404	26/01/26	Position paper on UNSC reform and peacekeeping operations.	90	104	20
5	405	27/01/26	Position paper on global pandemic preparedness.	94	105	30
6	406	27/01/26	Position paper on mental health access and policy.	87	106	30

attendance

	D_ID	SESSION_ID	STATUS
1	101	1001	present_and_voting
2	101	1002	present
3	101	1003	late
4	102	1001	present
5	102	1002	present_and_voting
6	102	1003	present
7	103	1001	present
8	103	1002	absent
9	103	1003	present
10	104	1001	late
11	104	1002	present
12	104	1003	present_and_voting
13	105	1001	present
14	105	1002	present
15	105	1003	present_and_voting
16	106	1001	absent
17	106	1002	present
18	106	1003	late

workshop_attendees

	D_ID	WORKSHOP_ID
1	101	501
2	101	502
3	102	501
4	103	502
5	104	503
6	105	501
7	105	503
8	106	502

award_recipients

	D_ID	AW_ID
1	101	301
2	101	302
3	102	302
4	103	303
5	104	303
6	105	304

topic

	TOPIC	COM_ID
1	Cybersecurity and international peace	20
2	Forced labor in global supply chains	10
3	Global pandemic preparedness	30
4	Mental health access	30
5	Peacekeeping reform	20
6	Youth unemployment in MENA	10

evaluations

	D_ID	CHAIR_ID	COM_ID
1	101	201	10
2	101	202	10
3	102	201	10
4	103	203	20
5	104	204	20
6	105	205	30
7	106	206	30

feedback

	D_ID	CHAIR_ID	COM_ID	PROFESSIONALISM	RESEARCH	P_SPEAKING
1	101	201	10	9	9	8
2	101	202	10	8	9	9
3	102	201	10	8	8	7
4	103	203	20	7	8	8
5	104	204	20	9	8	9
6	105	205	30	10	9	9
7	106	206	30	7	7	8

A sample of how we populated the tables is provided below. More information on how we populated all tables is provided in our full SQL script that is attached.

```
-- COUNTRY
INSERT INTO country VALUES (1, 'Middle East', 'United Arab Emirates', 'N');
INSERT INTO country VALUES (2, 'North America', 'United States of America', 'Y');
INSERT INTO country VALUES (3, 'Europe', 'France', 'Y');
INSERT INTO country VALUES (4, 'Africa', 'Egypt', 'N');
INSERT INTO country VALUES (5, 'Asia', 'India', 'N');
INSERT INTO country VALUES (6, 'Asia', 'China', 'Y');

-- COMMITTEE
INSERT INTO committee VALUES (10, 'ECOSOC', 'Specialized', 'ESB', '1001');
INSERT INTO committee VALUES (20, 'UNSC', 'Security Council', 'ESB', '1002');
INSERT INTO committee VALUES (30, 'WHO', 'Specialized', 'ESB', '1003');

-- TOPIC
INSERT INTO topic VALUES ('Youth unemployment in MENA', 10);
INSERT INTO topic VALUES ('Forced labor in global supply chains', 10);
INSERT INTO topic VALUES ('Cybersecurity and international peace', 20);
INSERT INTO topic VALUES ('Peacekeeping reform', 20);
INSERT INTO topic VALUES ('Global pandemic preparedness', 30);
INSERT INTO topic VALUES ('Mental health access', 30);
```

6.5 Sample Queries

The following 12 queries were run against the populated schema to validate the design.

Query 1: Show all delegates with their country and committee.

```
SELECT
    d.d_id,
    d.f_name || ' ' || d.l_name AS delegate_name,
    c.country_name,
    cm.com_name,
    d.regi_status
FROM delegate d
JOIN country c ON d.country_id = c.country_id
JOIN committee cm ON d.com_id = cm.com_id
ORDER BY d.d_id;
```

	D_ID	DELEGATE_NAME	COUNTRY_NAME	COM_NAME	REGI_STATUS
1	101	Ahmed Mohsen	Egypt	ECOSOC	Registered
2	102	Fedaa Elmahdi	United Arab Emirates	ECOSOC	Registered
3	103	Daniel Zakhari	United States of America	UNSC	Registered
4	104	Renad Hamad	France	UNSC	Registered
5	105	Sara Khan	India	WHO	Registered
6	106	Omar Li	China	WHO	Registered

Query 2: Count delegates in each committee.

```
SELECT
    cm.com_name,
    COUNT(d.d_id) AS number_of_delegates
FROM committee cm
LEFT JOIN delegate d ON cm.com_id = d.com_id
GROUP BY cm.com_name
ORDER BY cm.com_name;
```

	COM_NAME	NUMBER_OF_DELEGATES
1	ECOSOC	2
2	UNSC	2
3	WHO	2

Query 3: Show committee topics.

```
SELECT
    cm.com_name,
    t.topic
FROM committee cm
JOIN topic t ON cm.com_id = t.com_id
ORDER BY cm.com_name, t.topic;
```

	COM_NAME	TOPIC
1	ECOSOC	Forced labor in global supply chains
2	ECOSOC	Youth unemployment in MENA
3	UNSC	Cybersecurity and international peace
4	UNSC	Peacekeeping reform
5	WHO	Global pandemic preparedness
6	WHO	Mental health access

Query 4: Show attendance records with delegate names and session dates.

```
SELECT
    d.f_name || ' ' || d.l_name AS delegate_name,
    s.session_date,
    s.s_starttime,
    s.s_endtime,
    a.status
FROM attendance a
JOIN delegate d ON a.d_id = d.d_id
JOIN mun_session s ON a.session_id = s.session_id
ORDER BY d.d_id, s.session_id;
```

	DELEGATE_NAME	SESSION_DATE	S_STARTTIME	S_ENDTIME	STATUS
1	Ahmed Mohsen	30/01/26	09:00	10:30	present_and_voting
2	Ahmed Mohsen	30/01/26	11:00	12:30	present
3	Ahmed Mohsen	31/01/26	09:00	10:30	late
4	Fedaa Elmahdi	30/01/26	09:00	10:30	present
5	Fedaa Elmahdi	30/01/26	11:00	12:30	present_and_voting
6	Fedaa Elmahdi	31/01/26	09:00	10:30	present
7	Daniel Zakhari	30/01/26	09:00	10:30	present
8	Daniel Zakhari	30/01/26	11:00	12:30	absent
9	Daniel Zakhari	31/01/26	09:00	10:30	present
10	Renad Hamad	30/01/26	09:00	10:30	late
11	Renad Hamad	30/01/26	11:00	12:30	present
12	Renad Hamad	31/01/26	09:00	10:30	present_and_voting
13	Sara Khan	30/01/26	09:00	10:30	present
14	Sara Khan	30/01/26	11:00	12:30	present
15	Sara Khan	31/01/26	09:00	10:30	present_and_voting
16	Omar Li	30/01/26	09:00	10:30	absent
17	Omar Li	30/01/26	11:00	12:30	present
18	Omar Li	31/01/26	09:00	10:30	late

Query 5: Attendance summary per delegate.

```
SELECT
    d.f_name || ' ' || d.l_name AS delegate_name,
    SUM(CASE WHEN a.status IN ('present', 'present_and_voting') THEN 1 ELSE 0 END) AS
sessions_present,
    SUM(CASE WHEN a.status = 'late' THEN 1 ELSE 0 END) AS sessions_late,
    SUM(CASE WHEN a.status = 'absent' THEN 1 ELSE 0 END) AS sessions_absent
FROM delegate d
LEFT JOIN attendance a ON d.d_id = a.d_id
GROUP BY d.f_name, d.l_name
ORDER BY delegate_name;
```

	DELEGATE_NAME	SESSIONS_PRESENT	SESSIONS_LATE	SESSIONS_ABSENT
1	Ahmed Mohsen	2	1	0
2	Daniel Zakhari	2	0	1
3	Fedaa Elmahdi	3	0	0
4	Omar Li	1	1	1
5	Renad Hamad	2	1	0
6	Sara Khan	3	0	0

Query 6: Show position papers with scores.

```
SELECT
    pp.paper_no,
    d.f_name || ' ' || d.l_name AS delegate_name,
    cm.com_name,
    pp.sub_date,
    pp.score
FROM position_paper pp
JOIN delegate d ON pp.d_id = d.d_id
JOIN committee cm ON pp.com_id = cm.com_id
ORDER BY pp.score DESC;
```

	PAPER_NO	DELEGATE_NAME	COM_NAME	SUB_DATE	SCORE
1	405	Sara Khan	WHO	27/01/26	94
2	401	Ahmed Mohsen	ECOSOC	25/01/26	92
3	404	Renad Hamad	UNSC	26/01/26	90
4	402	Fedaa Elmahdi	ECOSOC	25/01/26	88
5	406	Omar Li	WHO	27/01/26	87
6	403	Daniel Zakhari	UNSC	26/01/26	85

Query 7: Show derived evaluation scores.

```
SELECT
    d.f_name || ' ' || d.l_name AS delegate_name,
    ch.f_name || ' ' || ch.l_name AS chair_name,
    cm.com_name,
    es.score_ev
FROM evaluation_scores es
JOIN delegate d ON es.d_id = d.d_id
JOIN chair ch ON es.chair_id = ch.chair_id
JOIN committee cm ON es.com_id = cm.com_id
ORDER BY es.score_ev DESC;
```

	DELEGATE_NAME	CHAIR_NAME	COM_NAME	SCORE_EV
1	Sara Khan	Mariam Saeed	WHO	9.33
2	Renad Hamad	Reema Shubair	UNSC	8.67
3	Ahmed Mohsen	Youssef Hassan	ECOSOC	8.67
4	Ahmed Mohsen	Ranim Ali	ECOSOC	8.67
5	Fedaa Elmahdi	Youssef Hassan	ECOSOC	7.67
6	Daniel Zakhari	Shamma Almualla	UNSC	7.67
7	Omar Li	Khaled Nasser	WHO	7.33

Query 8: Show award recipients.

```
SELECT
    aw.aw_title,
    cm.com_name,
    d.f_name || ' ' || d.l_name AS recipient_name,
    aw.a_date
FROM award_recipients ar
JOIN awards aw ON ar.aw_id = aw.aw_id
JOIN delegate d ON ar.d_id = d.d_id
JOIN committee cm ON aw.com_id = cm.com_id
ORDER BY aw.aw_title, recipient_name;
```

	AW_TITLE	COM_NAME	RECIPIENT_NAME	A_DATE
1	Best Delegate	ECOSOC	Ahmed Mohsen	01/02/26
2	Best Position Paper	WHO	Sara Khan	01/02/26
3	Honorable Mention	UNSC	Daniel Zakhari	01/02/26
4	Honorable Mention	UNSC	Renad Hamad	01/02/26
5	Outstanding Delegate	ECOSOC	Ahmed Mohsen	01/02/26
6	Outstanding Delegate	ECOSOC	Fedaa Elmahdi	01/02/26

Query 9: Show workshop attendees.

```
SELECT
    w.title AS workshop_title,
    w.speaker_name,
    d.f_name || ' ' || d.l_name AS delegate_name
FROM workshop_attendees wa
JOIN workshop w ON wa.workshop_id = w.workshop_id
JOIN delegate d ON wa.d_id = d.d_id
ORDER BY w.title, delegate_name;
```

	WORKSHOP_TITLE	SPEAKER_NAME	DELEGATE_NAME
1	Public Speaking for Delegates	Youssef Hassan	Ahmed Mohsen
2	Public Speaking for Delegates	Youssef Hassan	Daniel Zakhari
3	Public Speaking for Delegates	Youssef Hassan	Omar Li
4	Research and Position Paper Writing	Ranim Ali	Renad Hamad
5	Research and Position Paper Writing	Ranim Ali	Sara Khan
6	Rules of Procedure Training	Dr. Mohamed Ibrahim	Ahmed Mohsen
7	Rules of Procedure Training	Dr. Mohamed Ibrahim	Fedaa Elmahdi
8	Rules of Procedure Training	Dr. Mohamed Ibrahim	Sara Khan

Query 10: Verify each committee has at least two chairs.

```
SELECT
    cm.com_name,
    COUNT(ch.chair_id) AS number_of_chairs
FROM committee cm
LEFT JOIN chair ch ON cm.com_id = ch.com_id
GROUP BY cm.com_name
HAVING COUNT(ch.chair_id) >= 2
ORDER BY cm.com_name;
```

	COM_NAME	NUMBER_OF_CHAIRS
1	ECOSOC	2
2	UNSC	2
3	WHO	2

Query 11: Overall delegate performance using paper score and evaluation score.

```
SELECT
    d.d_id,
    d.f_name || ' ' || d.l_name AS delegate_name,
    cm.com_name,
    pp.score AS paper_score,
    ROUND(AVG(es.score_ev), 2) AS avg_evaluation_score,
    ROUND((pp.score / 10 + AVG(es.score_ev)) / 2, 2) AS overall_score
FROM delegate d
JOIN committee cm ON d.com_id = cm.com_id
JOIN position_paper pp ON d.d_id = pp.d_id
LEFT JOIN evaluation_scores es ON d.d_id = es.d_id AND d.com_id = es.com_id
GROUP BY d.d_id, d.f_name, d.l_name, cm.com_name, pp.score
ORDER BY overall_score DESC;
```

	D_ID	DELEGATE_NAME	COM_NAME	PAPER_SCORE	AVG_EVALUATION_SCORE	OVERALL_SCORE
1	105	Sara Khan	WHO	94	9.33	9.37
2	101	Ahmed Mohsen	ECOSOC	92	8.67	8.94
3	104	Renad Hamad	UNSC	90	8.67	8.84
4	102	Fedaa Elmahdi	ECOSOC	88	7.67	8.24
5	103	Daniel Zakhari	UNSC	85	7.67	8.09
6	106	Omar Li	WHO	87	7.33	8.02

Query 12: Show full feedback components for each evaluation.

SELECT

```

d.f_name || ' ' || d.l_name AS delegate_name,
ch.f_name || ' ' || ch.l_name AS chair_name,
cm.com_name,
f.professionalism,
f.research,
f.p_speaking,
ROUND((f.professionalism + f.research + f.p_speaking) / 3, 2) AS derived_score

```

FROM feedback f

JOIN delegate d ON f.d_id = d.d_id

JOIN chair ch ON f.chair_id = ch.chair_id

JOIN committee cm ON f.com_id = cm.com_id

ORDER BY derived_score DESC;

	DELEGATE_NAME	CHAIR_NAME	COM_NAME	PROFESSIONALISM	RESEARCH	P_SPEAKING	DERIVED_SCORE
1	Sara Khan	Mariam Saeed	WHO	10	9	9	9.33
2	Renad Hamad	Reema Shubair	UNSC	9	8	9	8.67
3	Ahmed Mohsen	Youssef Hassan	ECOSOC	9	9	8	8.67
4	Ahmed Mohsen	Ranim Ali	ECOSOC	8	9	9	8.67
5	Fedaa Elmahdi	Youssef Hassan	ECOSOC	8	8	7	7.67
6	Daniel Zakhari	Shamma Almualla	UNSC	7	8	8	7.67
7	Omar Li	Khaled Nasser	WHO	7	7	8	7.33

7. GUI Design in Apache Netbeans (Java)

To provide a user-friendly and efficient method for interacting with the AUSMUN Management System, a desktop application was developed using Java Swing and integrated with the backend Oracle database via JDBC (Java Database Connectivity). The interface functions as an administrative dashboard, allowing users to execute full CRUD (Create, Read, Update, Delete) and SQL query operations such as search across delegates, committees, attendance records, and position papers without requiring direct SQL command-line interactions. The program also allows them to generate reports utilizing SQL queries & subqueries for:

- All Delegates
- Attendance Details
- Evaluation Scores
- Delegates Per Committee
- Attendance Summary
- Committee Topics
- Award Recipients
- Position Paper Scores

7.1. Client-Side Domain Constraints

While the Oracle database layer enforces strict relational integrity through primary keys, foreign keys, and check constraints, relying solely on database exceptions for error handling leads to a poor user experience and unnecessary network overhead. Therefore, the GUI incorporates robust client-side data validation within *MainFrame.java* to intercept invalid data before an insert or update transaction is committed to the database.

Form Field / Component	Exact Implementation / Constraint
Email Address Field (emailField)	Passes input text through the isValidEmail() validation method, which matches the string against the regular expression pattern: ^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\$
Date Fields (dobField, submissionDateField)	Passes input text through the isValidDate() validation method, which applies a strict

	case-insensitive DateFormatter configured with the English pattern: dd-MMM-yyyy.
Numeric Attribute Fields (scoreField, paperNumberField, ID inputs)	Intercepts input fields by performing explicit data conversion testing. If the string fails conversion checks, a validation error dialog alerts the user that the field must contain a number.
Mandatory Text Inputs (Names, Titles, and Content Areas)	Enforces presence checks by stripping leading/trailing whitespace via string-trimming operations and verifying the field is not empty (<code>.getText().trim().isEmpty()</code>).
Relational Input Selections (Committees, Countries, and Delegates ComboBoxes)	Restricts user selections to predefined dropdown options loaded straight from the base lookup tables via <code>loadCountries()</code> , <code>loadCommittees()</code> , and <code>loadDelegates()</code> .

7.2. GUI Interface & Functionality

1. Committee

The screenshot displays the 'MUN Conference Database Management System' window. At the top, there are tabs for 'Committees', 'Delegates', 'Attendance', 'Reports', and 'Position Papers'. The 'Committees' tab is active. Below the tabs, there are five input fields labeled 'Committee ID', 'Committee Name', 'Committee Type', 'Building Name', and 'Room Number'. Below these fields are six buttons: 'Add', 'Update', 'Delete', 'Search', 'Clear', and 'Refresh'. At the bottom, there is a table with the following data:

Committee ID	Committee Name	Committee Type	Building Name	Room Number
10	ECOSOC	Specialized	ESB	1001
20	UNSC	Security Council	ESB	1002
30	WHO	Specialized	ESB	1003

As can be seen, the GUI allows full CRUD (Create, Read, Update, Delete) capabilities. Here is an example of a new Committee being created via the GUI:

MUN Conference Database Management System

Committees Delegates Attendance Reports Position Papers

Committee ID: 40
 Committee Name: UNGA
 Committee Type: General Assembly
 Building Name: ESB
 Room Number: 1004

Add Update Delete Search Clear Refresh

Committee ID	Committee Name	Committee Type	Building Name	Room Number
10	ECOSOC	Specialized	ESB	1001
20	UNSC	Security Council	ESB	1002
30	WHO	Specialized	ESB	1003
40	UNGA	General Assembly	ESB	1004

2. Delegates

MUN Conference Database Management System

Committees Delegates Attendance Reports Position Papers

Delegate ID: 103
 First Name: Daniel
 Last Name: Zakhari
 Email: daniel.zakhari@example.com
 School: American University of Sharjah
 Date of Birth: 11-DEC-2006
 Status: Registered
 Country: United States of America
 Committee: UNSC

Add Update Delete Search Clear Refresh

Delegate ID	First Name	Last Name	Email	School	Date of Birth	Status	Country	Committee
101	Ahmed	Mohsen	ahmed.moh...	American U...	15-MAR-2...	Registered	Egypt	ECOSOC
102	Fedaa	Elmahdi	fedaa.elma...	American U...	20-OCT-20...	Registered	United Arab...	ECOSOC
103	Daniel	Zakhari	daniel.zakh...	American U...	11-DEC-20...	Registered	United State...	UNSC
104	Renad	Hamad	renad.hama...	American U...	09-JUL-2004	Registered	France	UNSC
105	Sara	Khan	sara.khan@...	AUS High Sc...	11-MAY-20...	Registered	India	WHO
106	Omar	Li	omar.li@ex...	Sharjah Inte...	02-DEC-20...	Registered	China	WHO

Here is an example of a Delegate's information being updated:

MUN Conference Database Management System

Committees Delegates Attendance Reports Position Papers

Delegate ID: 103
 First Name: Daniel
 Last Name: Zakhari
 Email: danieltareks@gmail.com
 School: American University of Shariah
 Date of Birth: 11-DEC-2006
 Status: Registered
 Country: United States of America
 Committee: UNSC

Add Update Delete Search Clear Refresh

Delegate ID	First Name	Last Name	Email	School	Date of Birth	Status	Country	Committee
101	Ahmed	Mohsen	ahmed.moh...	American U...	15-MAR-2...	Registered	Egypt	ECOSOC
102	Fedaa	Elmahdi	fedaa.elma...	American U...	20-OCT-20...	Registered	United Arab...	ECOSOC
103	Daniel	Zakhari	danieltarek...	American U...	11-DEC-20...	Registered	United State...	UNSC
104	Renad	Hamad	renad.hama...	American U...	09-JUL-2004	Registered	France	UNSC
105	Sara	Khan	sara.khan@...	AUS High Sc...	11-MAY-20...	Registered	India	WHO
106	Omar	Li	omar.li@ex...	Sharjah Inte...	02-DEC-20...	Registered	China	WHO

3. Attendance

MUN Conference Database Management System

Committees Delegates Attendance Reports Position Papers

Delegate: 101 - Ahmed Mohsen
 Session: 1001 - 30-JAN-2026 09:00-10:30
 Status: present

Add Update Delete Search Clear Refresh

Delegate ID	Delegate Name	Session ID	Session Date	Start Time	End Time	Status
101	Ahmed Mohsen	1001	30-JAN-2026	09:00	10:30	present_and_voting
101	Ahmed Mohsen	1002	30-JAN-2026	11:00	12:30	present
101	Ahmed Mohsen	1003	31-JAN-2026	09:00	10:30	late
102	Fedaa Elmahdi	1001	30-JAN-2026	09:00	10:30	present
102	Fedaa Elmahdi	1002	30-JAN-2026	11:00	12:30	present_and_voting
102	Fedaa Elmahdi	1003	31-JAN-2026	09:00	10:30	present
103	Daniel Zakhari	1001	30-JAN-2026	09:00	10:30	present
103	Daniel Zakhari	1002	30-JAN-2026	11:00	12:30	absent
103	Daniel Zakhari	1003	31-JAN-2026	09:00	10:30	present
104	Renad Hamad	1001	30-JAN-2026	09:00	10:30	late
104	Renad Hamad	1002	30-JAN-2026	11:00	12:30	present
104	Renad Hamad	1003	31-JAN-2026	09:00	10:30	present_and_voting
105	Sara Khan	1001	30-JAN-2026	09:00	10:30	present
105	Sara Khan	1002	30-JAN-2026	11:00	12:30	present
105	Sara Khan	1003	31-JAN-2026	09:00	10:30	present_and_voting
106	Omar Li	1001	30-JAN-2026	09:00	10:30	absent
106	Omar Li	1002	30-JAN-2026	11:00	12:30	present
106	Omar Li	1003	31-JAN-2026	09:00	10:30	late

Here is an example of a Delegate's attendance record for a specific session being searched for using a query:

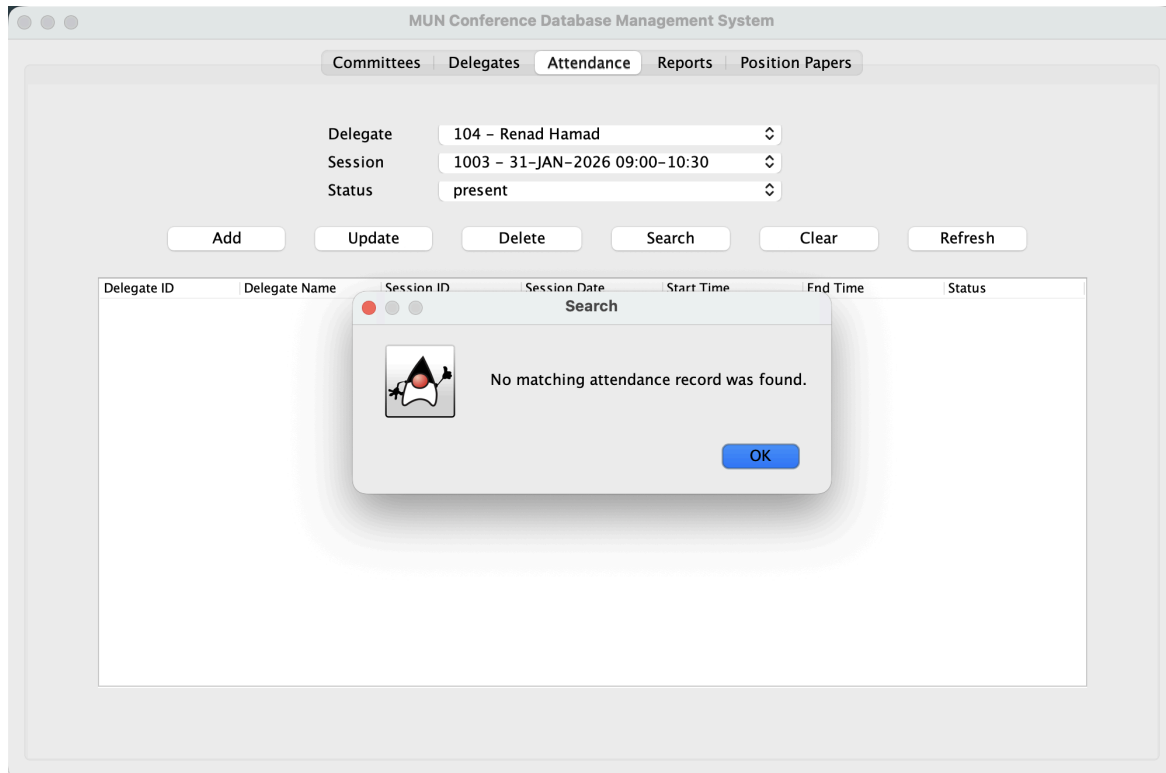
The screenshot shows the 'MUN Conference Database Management System' interface. The 'Attendance' tab is selected. The search criteria are: Delegate: 101 - Ahmed Mohsen, Session: 1002 - 30-JAN-2026 11:00-12:30, Status: present. The results table shows one record for Delegate ID 101, Delegate Name Ahmed Mohsen, Session ID 1002, Session Date 30-JAN-2026, Start Time 11:00, End Time 12:30, and Status present.

Delegate ID	Delegate Name	Session ID	Session Date	Start Time	End Time	Status
101	Ahmed Mohsen	1002	30-JAN-2026	11:00	12:30	present

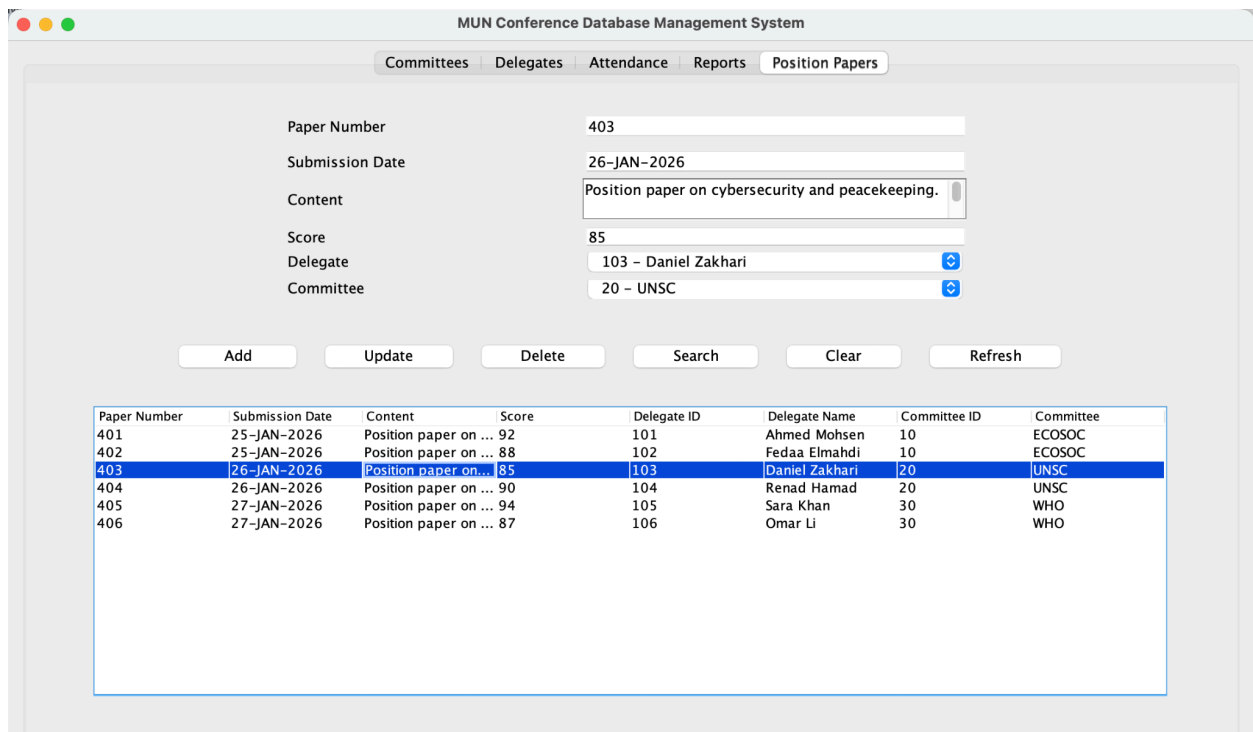
Here is an example of an attendance record being deleted entirely:

The screenshot shows the 'MUN Conference Database Management System' interface. The 'Attendance' tab is selected. The search criteria are: Delegate: 104 - Renad Hamad, Session: 1003 - 31-JAN-2026 09:00-10:30, Status: present. A 'Confirm Delete' dialog box is displayed over the results table, asking 'Delete this attendance record?' with 'No' and 'Yes' buttons. The results table shows one record for Delegate ID 104, Delegate Name Renad Hamad, Session ID 10, Session Date 31-JAN-2026, Start Time 09:00, End Time 10:30, and Status present_and_voting.

Delegate ID	Delegate Name	Session ID	Session Date	Start Time	End Time	Status
104	Renad Hamad	10	31-JAN-2026	09:00	10:30	present_and_voting

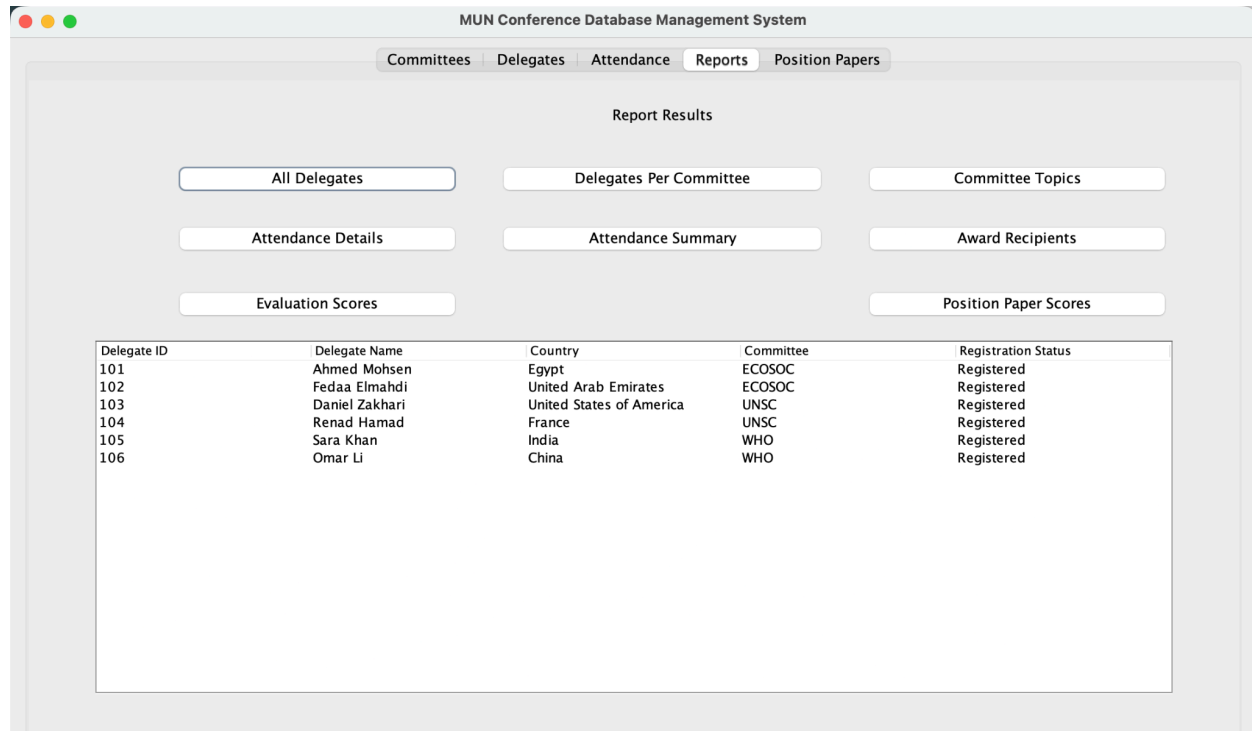


4. Position Papers



5. Reports

Reports can be generated for administrative purposes using SQL queries:



MUN Conference Database Management System

Committees Delegates Attendance **Reports** Position Papers

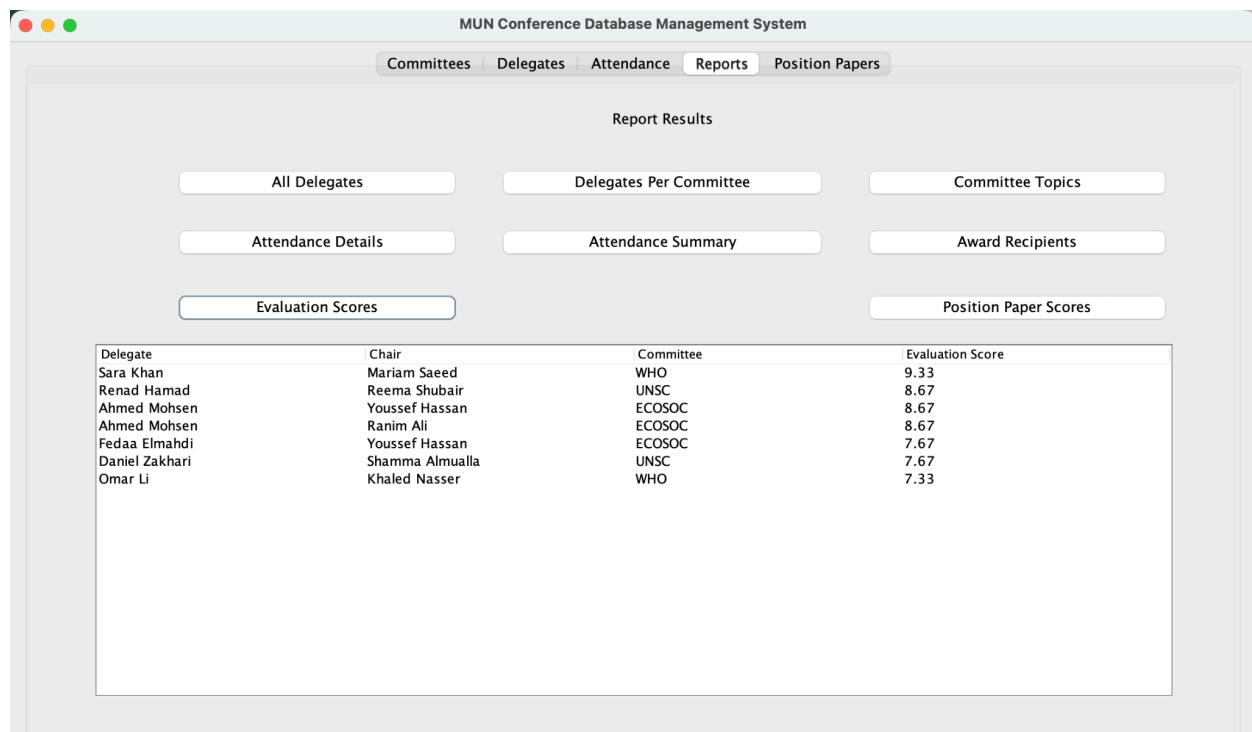
Report Results

All Delegates Delegates Per Committee Committee Topics

Attendance Details Attendance Summary Award Recipients

Evaluation Scores Position Paper Scores

Delegate ID	Delegate Name	Country	Committee	Registration Status
101	Ahmed Mohsen	Egypt	ECOSOC	Registered
102	Fedaa Elmahdi	United Arab Emirates	ECOSOC	Registered
103	Daniel Zakhari	United States of America	UNSC	Registered
104	Renad Hamad	France	UNSC	Registered
105	Sara Khan	India	WHO	Registered
106	Omar Li	China	WHO	Registered



MUN Conference Database Management System

Committees Delegates Attendance **Reports** Position Papers

Report Results

All Delegates Delegates Per Committee Committee Topics

Attendance Details Attendance Summary Award Recipients

Evaluation Scores Position Paper Scores

Delegate	Chair	Committee	Evaluation Score
Sara Khan	Mariam Saeed	WHO	9.33
Renad Hamad	Reema Shubair	UNSC	8.67
Ahmed Mohsen	Youssef Hassan	ECOSOC	8.67
Ahmed Mohsen	Ranim Ali	ECOSOC	8.67
Fedaa Elmahdi	Youssef Hassan	ECOSOC	7.67
Daniel Zakhari	Shamma Almualla	UNSC	7.67
Omar Li	Khaled Nasser	WHO	7.33

6. Miscellaneous Features (Resizing)

We ensured that our GUI could fit on all screens by ensuring resizing capability:

Small View (880 x 560):

The screenshot shows the 'MUN Conference Database Management System' window in a small view. The window has a title bar with standard macOS window controls. Below the title bar is a navigation bar with tabs: 'Committees', 'Delegates', 'Attendance', 'Reports', and 'Position Papers'. The 'Committees' tab is selected. The main content area contains a form with five input fields: 'Committee ID', 'Committee Name', 'Committee Type', 'Building Name', and 'Room Number'. Below the form are six buttons: 'Add', 'Update', 'Delete', 'Search', 'Clear', and 'Refresh'. At the bottom is a table with five columns: 'Committee ID', 'Committee Name', 'Committee Type', 'Building Name', and 'Room Number'. The table contains four rows of data.

Committee ID	Committee Name	Committee Type	Building Name	Room Number
10	ECOSOC	Specialized	ESB	1001
20	UNSC	Security Council	ESB	1002
30	WHO	Specialized	ESB	1003
40	UNGA	General Assembly	ESB	1004

Fullscreen View (1470 x 870):

The screenshot shows the 'MUN Conference Database Management System' window in a fullscreen view. The window is larger, and the content is more spread out. The navigation bar and form are still present, but the table is now wider and contains the same data as in the small view.

Committee ID	Committee Name	Committee Type	Building Name	Room Number
10	ECOSOC	Specialized	ESB	1001
20	UNSC	Security Council	ESB	1002
30	WHO	Specialized	ESB	1003
40	UNGA	General Assembly	ESB	1004

8. Conclusion

The development of the **AUSMUN Management System** successfully replaces traditional, fragmented spreadsheet tracking with a centralized, relationally sound database application engineered specifically for Model United Nations conferences at the American University of Sharjah. By systematically navigating the complete database design lifecycle, the project evolved from an abstract administrative problem into a fully functioning, end-to-end enterprise software solution.

Our team's development process closely adhered to rigorous software and database engineering methodologies:

- **Conceptual Modeling:** We initiated the cycle by constructing a comprehensive Entity-Relationship (ER) model consisting of eight strong entities and one weak entity. This layer uniquely integrated intricate real-world domain requirements, including composite names, multivalued topics, and a specialized ternary relationship (Evaluates) to capture committee-specific delegate grading.
- **Logical Relational Mapping:** The conceptual design was methodically transformed into a highly organized 14-table relational schema. During this phase, primary and foreign keys were assigned to map entities and resolve complex 1:1, 1:N, and M:N relationships into structurally secure tables.
- **Physical Implementation:** Realized inside an Oracle SQL schema, the physical data layer was hardened with critical domain rules, unique constraints, and check bounds. To handle Oracle database restrictions seamlessly, structural adaptations were integrated—such as renaming the reserved keyword Session to mun_session. Furthermore, a specialized database view (evaluation_scores) was built to dynamically compute derived performance scores across multi-dimensional feedback categories without introducing data redundancy.
- **Verification and Testing:** The backend was seeded with comprehensive mock data simulating an active, multi-committee conference scenario. A suite of 12 multi-table validation queries successfully tested the schema's relational integrity, verifying that everything from complex performance aggregations to attendance summaries operated accurately.
- **Frontend Integration:** Finally, we bridged the backend relational schema with a desktop-based Graphical User Interface (GUI) engineered in Apache NetBeans using Java Swing and JDBC. This frontend includes client-side regular expressions and strict datetime input validation to safeguard data integrity right at the point of entry, avoiding unneeded database exception overhead.

System Functionality & Impact

The final application delivers an administrative dashboard that abstractly handles direct command-line database interactions for end-users. Conference organizers and committee chairs are equipped with full CRUD (Create, Read, Update, Delete) capabilities to manage delegate registrations, country representations, committee assignments, session attendance, position papers, and award distributions smoothly. At the same time, the system features robust reporting mechanics, allowing users to extract instantaneous performance metrics, multi-variable summaries, and logistics logs at the click of a button.

Ultimately, the AUSMUN Management System stands as a scalable, high-performance solution that optimizes large-scale event operations, drastically reduces manual cross-referencing bottlenecks, and ensures absolute data integrity over the course of a conference.